DeltaGrad: Towards Robust Deep Learning via Adaptive Gradient Reliability

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1 Introduction

Modern adaptive optimizers like Adam [1] often "over-smooth" gradients, leading to noise memorization and instability. While SOTA methods like SAM [5] improve generalization by smoothing the loss landscape, they incur high computational costs. In contrast, **DeltaGrad** introduces a real-time *Reliability Metric* (R_t) to modulate updates via instantaneous gradient coherence. This allows DeltaGrad to "tame" instability on CIFAR-100 [3], maintaining convergence even at 10× the standard learning rate. With a negligible 0.38% overhead at a large batch size, DeltaGrad offers a responsive, efficient alternative for high-noise, non-convex optimization.

2 Methodology

DeltaGrad introduces a self-regulating update mechanism based on gradient coherence. First, a lightly smoothed gradient $g^{(t)}_{smooth}$ is computed:

$$g^{(t)}_{smooth} = \sigma g^{(t-1)}_{smooth} + (1-\sigma)\nabla L^{(t)} \tag{1}$$

using a low-range factor $\sigma \in [10^{-4}, 0.1]$ to ensure stability gains originate from reliability logic rather than damping. The core innovation is the **Reliability Metric (R_t)**:

$$R_t = \text{clamp}\left(\frac{K - \sum_{j=0}^{K-1} \alpha^j \Phi_j}{K}, 0.1, 1.0\right) \tag{2}$$

which quantifies the reliability of the signal against a window of $K-1$ previous gradients. Clamping prevents gradient paralysis. Computing a normalized distance Φ_j:

$$\Phi_j = \frac{|g^{(t)}_{smooth} - g^{(t-j)}|}{|g^{(t)}_{smooth}| + |g^{(t-j)}| + \epsilon} \tag{3}$$

Weighted by exponential decay α, R_t acts as a dynamic safety brake: it approaches 1.0 during coherent trajectories and scales down to 0.1 when noise dominates.

The update direction is governed by the **Windowed Inertia ($\hat{m}_t$)**:

$$\hat{m}_t = \frac{\sum_{j=0}^{K-1} \beta^{j+1} \cdot g^{(t-j)}}{\sum_{j=0}^{K-1} \beta^{j+1}} \tag{4}$$

Unlike infinite-horizon EMAs, this finite memory remains responsive to local geometry. The final update integrates these components:

$$\theta_{t+1} = \theta_t - (\mu \cdot \gamma) \cdot (\hat{m}_t \odot R_t) \tag{5}$$

where μ is the learning rate and γ is a sensitivity coefficient. The product $\hat{m}_t \odot R_t$ allows for fine-grained dampening, suppressing oscillations that lead to divergence.

3 Experiments

Three experiments were conducted to evaluate the robustness of DeltaGrad: a Learning Rate Stress Test, a Batch Size Test, and a Data Noise Test. To provide a consistent baseline, a compact ConvNet architecture was trained using both Adam and DeltaGrad, with both optimizers tuned via Optuna [4] for an equal number of trials on the CIFAR-100 dataset[3]. This extended abstract focuses on the results from the Data Noise and Learning Rate Stress Tests; further results and the complete implementation are available in the official repository [7].

3.1 General Results

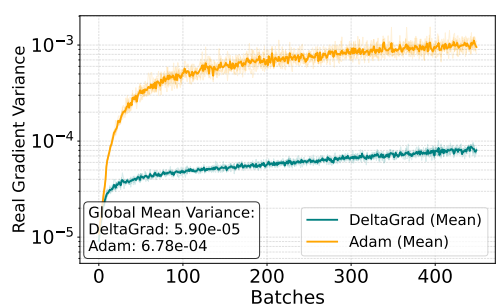
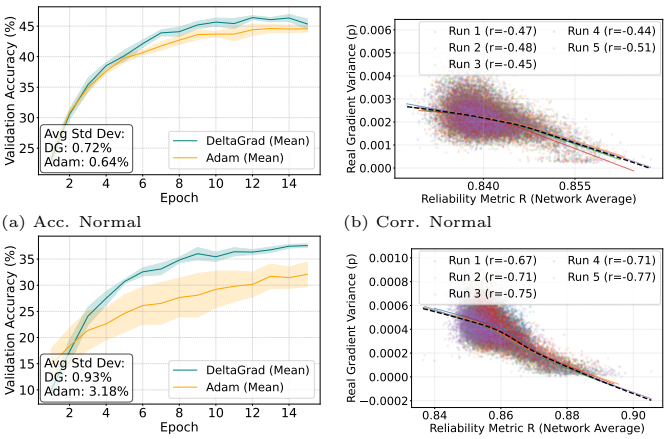


Figure 1: Gradient Variance Comparison for batch size of 512 over 5 runs. Mean(highlighted) ; Deviation(faint)

High gradient variance indicates that a neuron has not yet converged to a stable feature representation, often due to conflicting signals from labeled noise [6]. By prioritizing parameters that exhibit directional consistency, DeltaGrad effectively favors 'feature learning' over 'noise memorization'.

In all tests, DeltaGrad stabilizes gradient variance an order of magnitude below Adam [1], demonstrating that the reliability metric R (Eq. 2) effectively guides the optimizer toward more stable regions, resulting in more robust learning.

3.2 Learning Rate Stress Test



(a) Acc. Normal (b) Corr. Normal

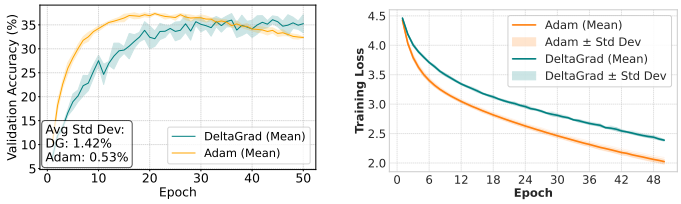
(e) Acc. 10x (f) Corr. 10x

Figure 2: Learning rate stress test (BS=16) across two regimes (Baseline, 10×). Shaded areas represent standard deviation over five runs. The plots show a negative correlation between the Reliability Metric (R) and gradient variance on all runs.

As illustrated in Figure 2, DeltaGrad exhibits significantly higher consistency in validation accuracy across varying learning rate regimes compared to Adam [1]. Using a fixed batch size of 16 to intentionally amplify stochastic noise, the results show that the correlation between gradient variance and the reliability factor R strengthens with elevated learning rates.

Interestingly, despite the increased noise typical of higher learning rates, DeltaGrad maintains lower gradient variance in these regimes. This is accompanied by a normal accuracy learning curve that eliminates the gradient paralysis scenario [6]. These findings suggest that R functions as an effective damping mechanism, suppressing the high-frequency oscillations common in noisy landscapes [5] and ensuring stability even under aggressive optimization settings.

3.3 Data Noise Test



(a) Acc. Normal (b) Corr. Normal

Figure 3: Accuracy and Training Loss under 20% label noise (BS=512).

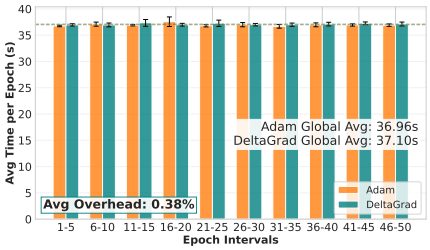


Figure 4: Mean training time per epoch (BS=512, 20% noise). DeltaGrad introduces a negligible computational overhead of only 0.38% compared to Adam.

Under 20% label noise (BS=512), DeltaGrad demonstrates superior robustness compared to Adam [1]. While Adam suffers from accuracy degradation after epoch 20 due to noise memorization - evidenced by its lower training loss - DeltaGrad maintains a stable upward trajectory, reaching a higher final accuracy. The higher training loss in DeltaGrad confirms the efficacy of the R metric in filtering inconsistent signals, prioritizing generalization over the minimization of noisy objectives [6]. This resilience is achieved with a negligible computational overhead of only 0.38%, making the mechanism potentially scalable for production environments.

Increased accuracy deviation at $BS = 512$ stems from reduced R-update granularity and the heightened impact of Windowed Inertia ($\hat{m}_t$) over fewer stochastic steps.

4 Conclusions

While these preliminary results confirm DeltaGrad's core mechanics, future work must overcome computational constraints to evaluate the Reliability Metric (Eq. 2) across a broader range of datasets and neural architectures, like Transformers [2]. These findings position DeltaGrad as a robust architectural foundation for noise-resilient optimization in high-stakes production environments.

References

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